

FROM POSSIBILITY TO ACTUALITY

ICT913 - Networking

LECTURE 8

Network Design and Implementation

CCNA 200-301, Volume I

Chapters 11-14

Perspectives on IPv4 Subnetting

Analyzing Classful IPv4 Networks

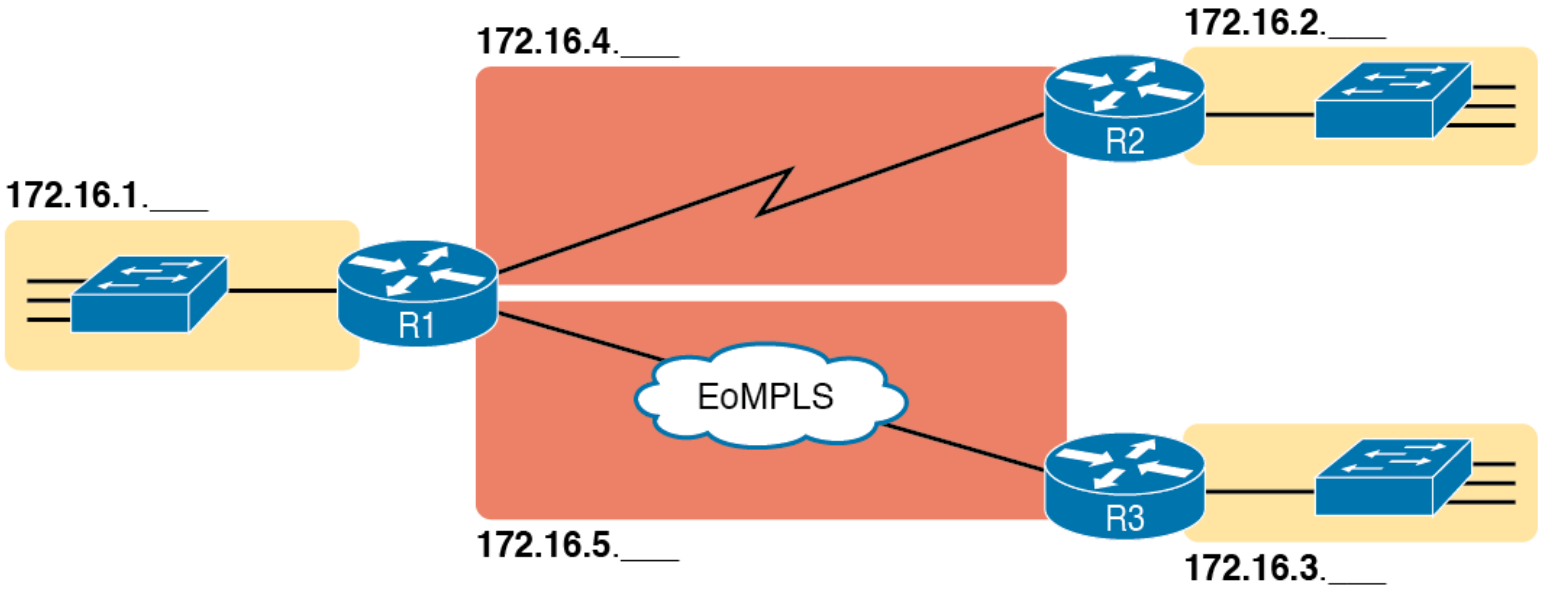
Analyzing Subnet Masks

Analyzing Existing Subnets

Objectives

- Analyze Requirements
- Make Design Choices
- Classful Network Concepts
- Subnet Mask Conversion
- Defining the Format of IPv4 Addresses
- Defining a Subnet
- Analyzing Existing Subnets: Binary
- Analyzing Existing Subnets: Decimal

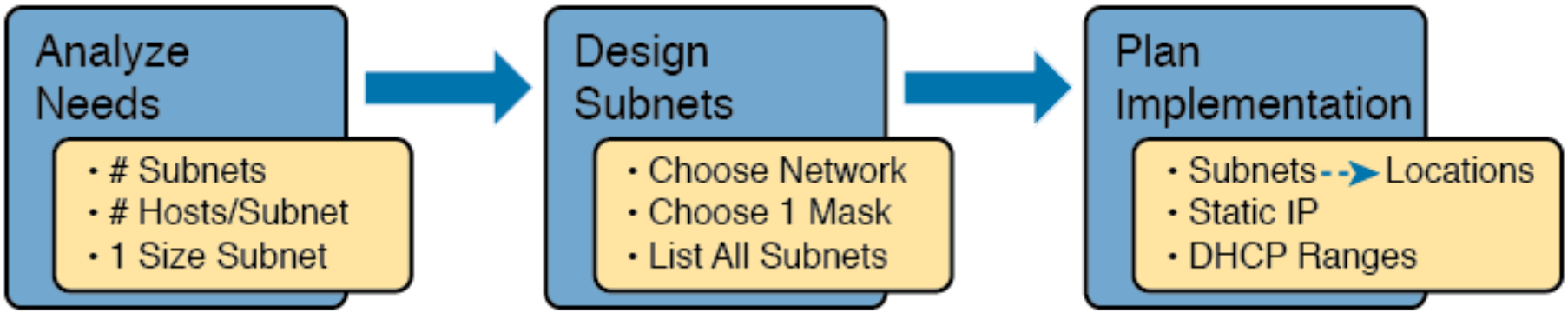
Example Subnet Plan Document



Subnet Design:

Class B 172.16.0.0
First 3 Octets are Equal

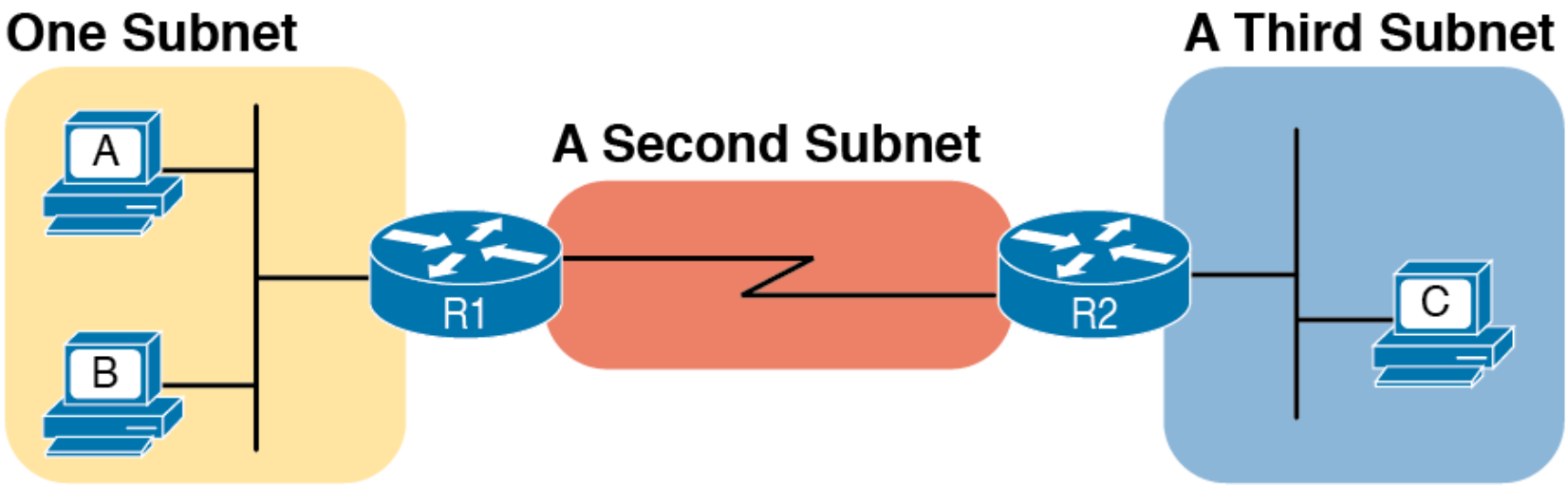
Subnet Planning, Design, and Implementation Tasks



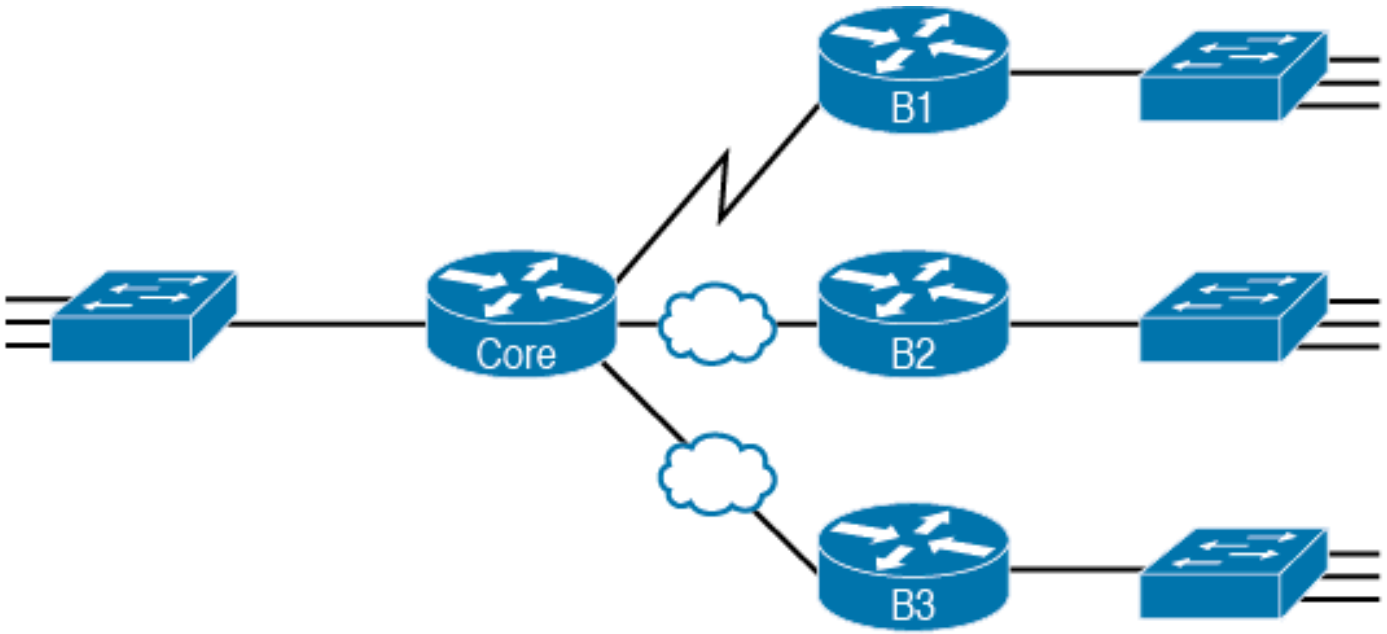
Analyze Subnetting and Addressing Needs

- Which hosts should be grouped together into a subnet?
- How many subnets does this network require?
- How many host IP addresses does each subnet require?
- Will we use a single subnet size for simplicity, or not?

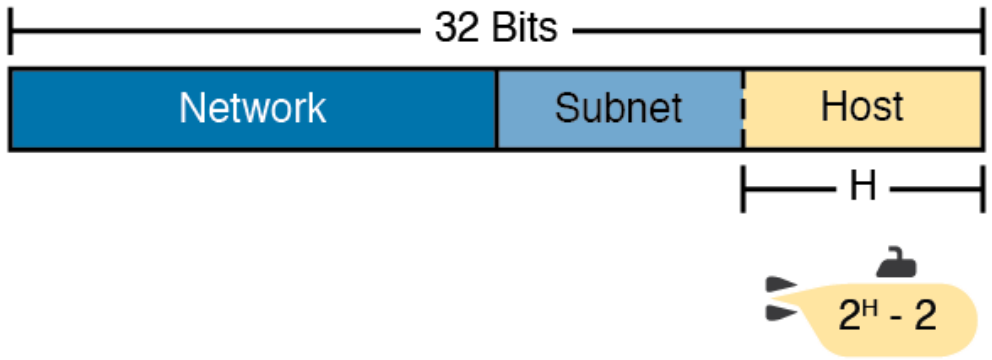
PC A and B in One Subnet, PC C in a Different Subnet



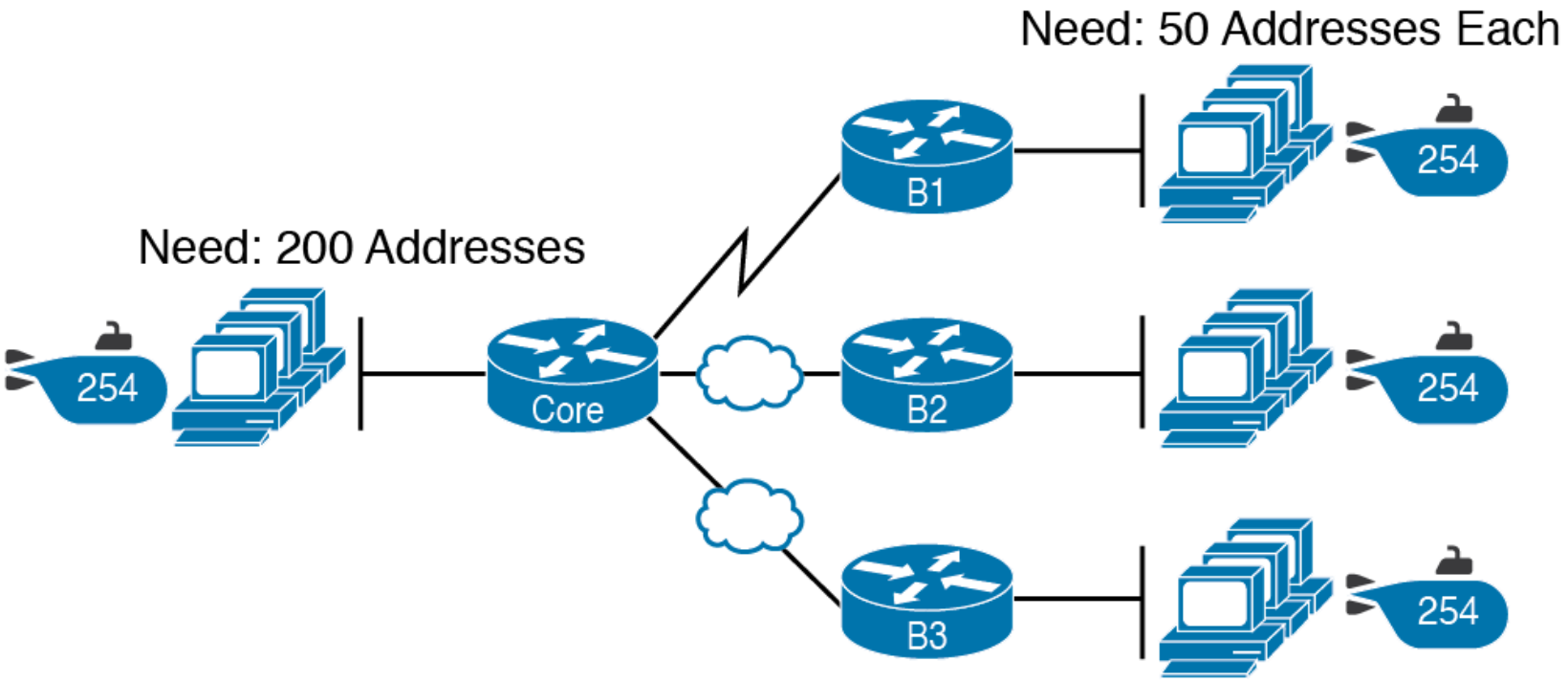
Four Site Internetwork with Small Central Site



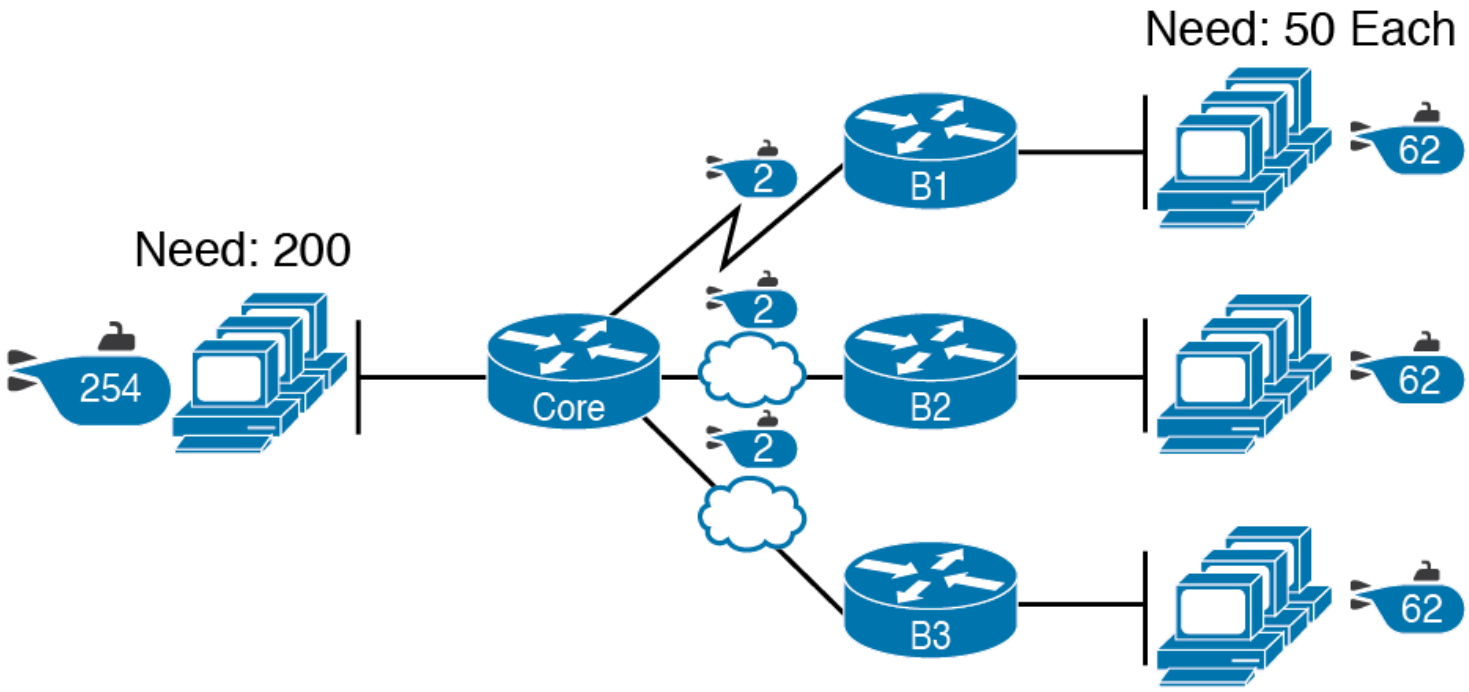
Subnet Size Concepts



Network Using One Subnet Size



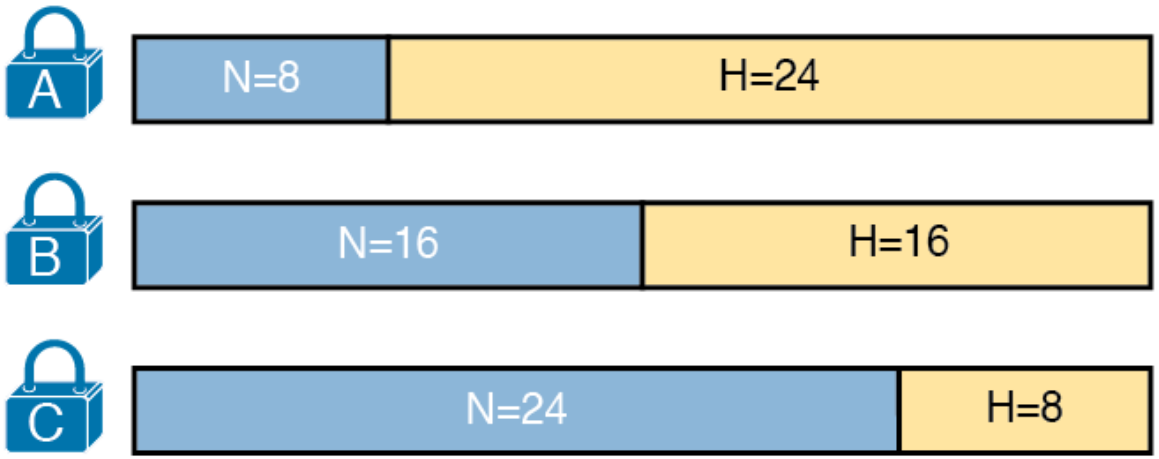
Three Masks, Three Subnet Sizes



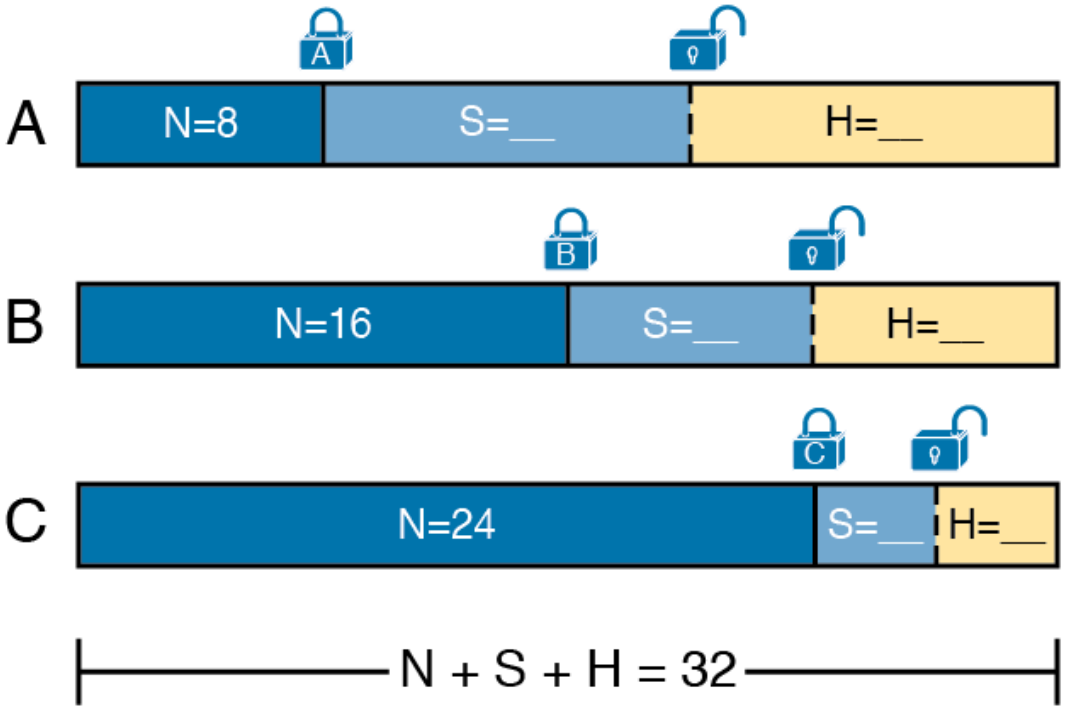
RFC 1918 Private Address Space

Private IP Networks	Class of Networks	Number of Networks
10.0.0.0	A	1
172.16.0.0 through 172.31.0.0	B	16
192.168.0.0 through 192.168.255.0	C	256

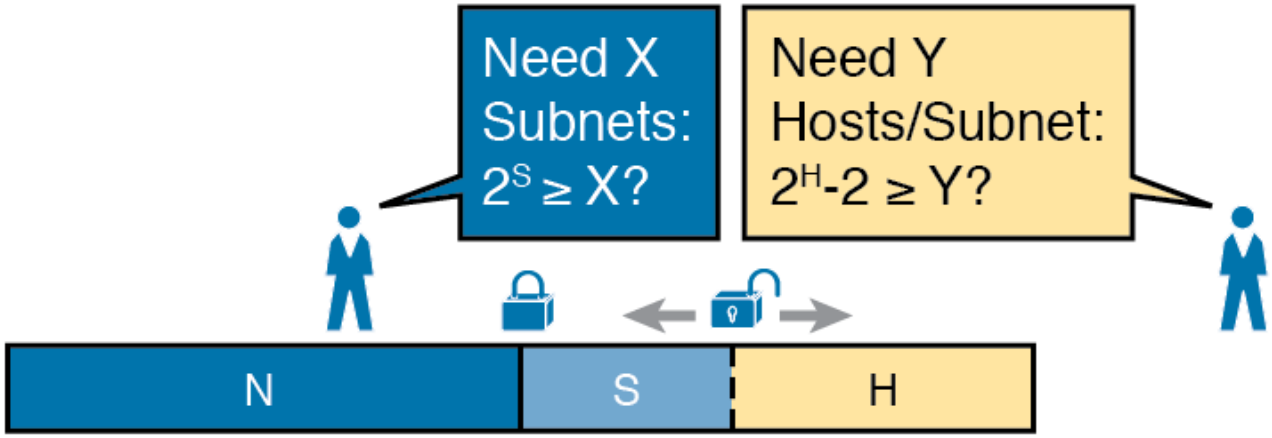
Format of Unsubnetted Class A, B, and C Networks



Concept of Borrowing Host Bits



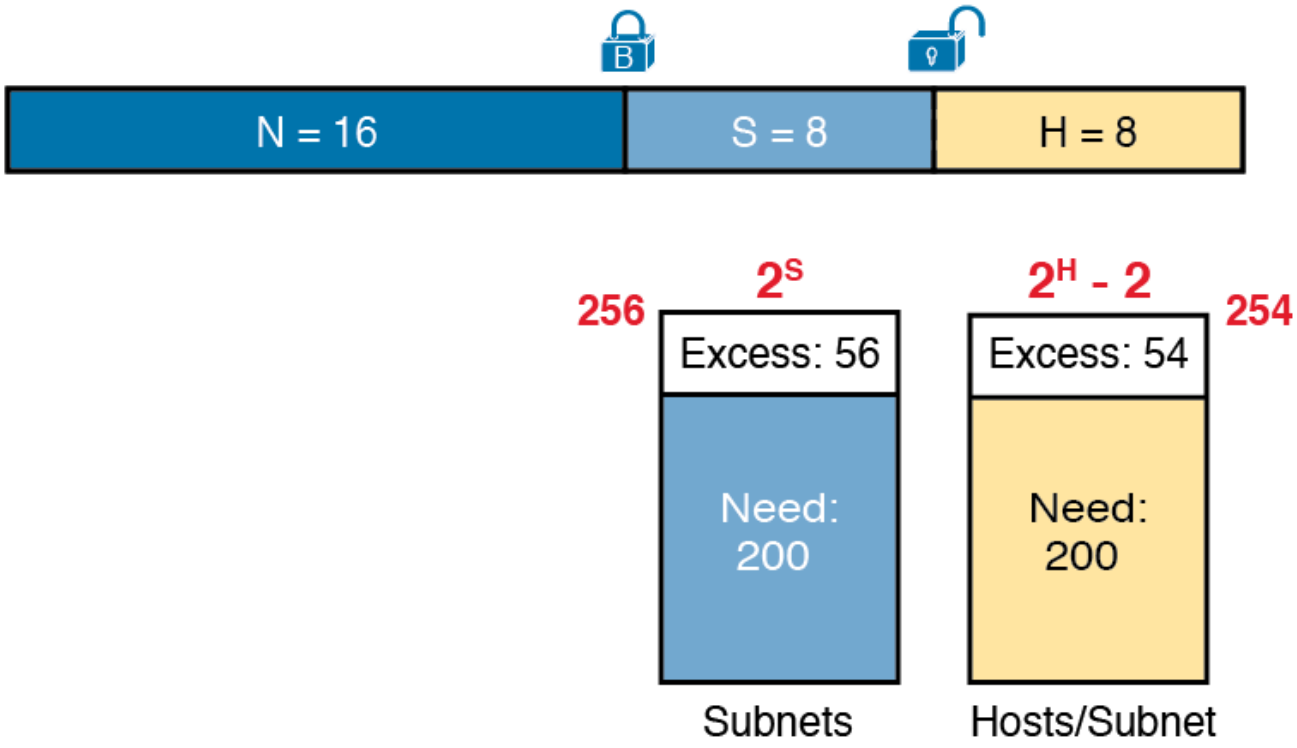
Borrowing Enough Subnet and Host Bits



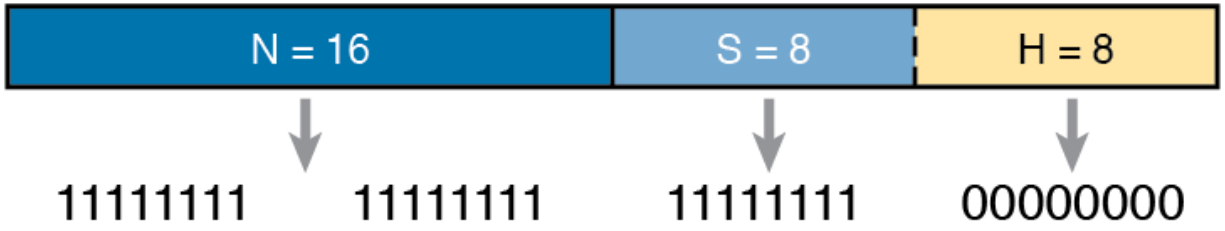
Powers of 2 Reference for Designing Masks

Number of Bits	2^X	$2^X - 2$
1	2	0
2	4	2
3	8	6
4	16	14
5	32	30
6	64	62
7	128	126
8	256	254
9	512	510
10	1024	1022
11	2048	2046
12	4096	4094

Example Mask Choice, N = 16, S = 8, H = 8



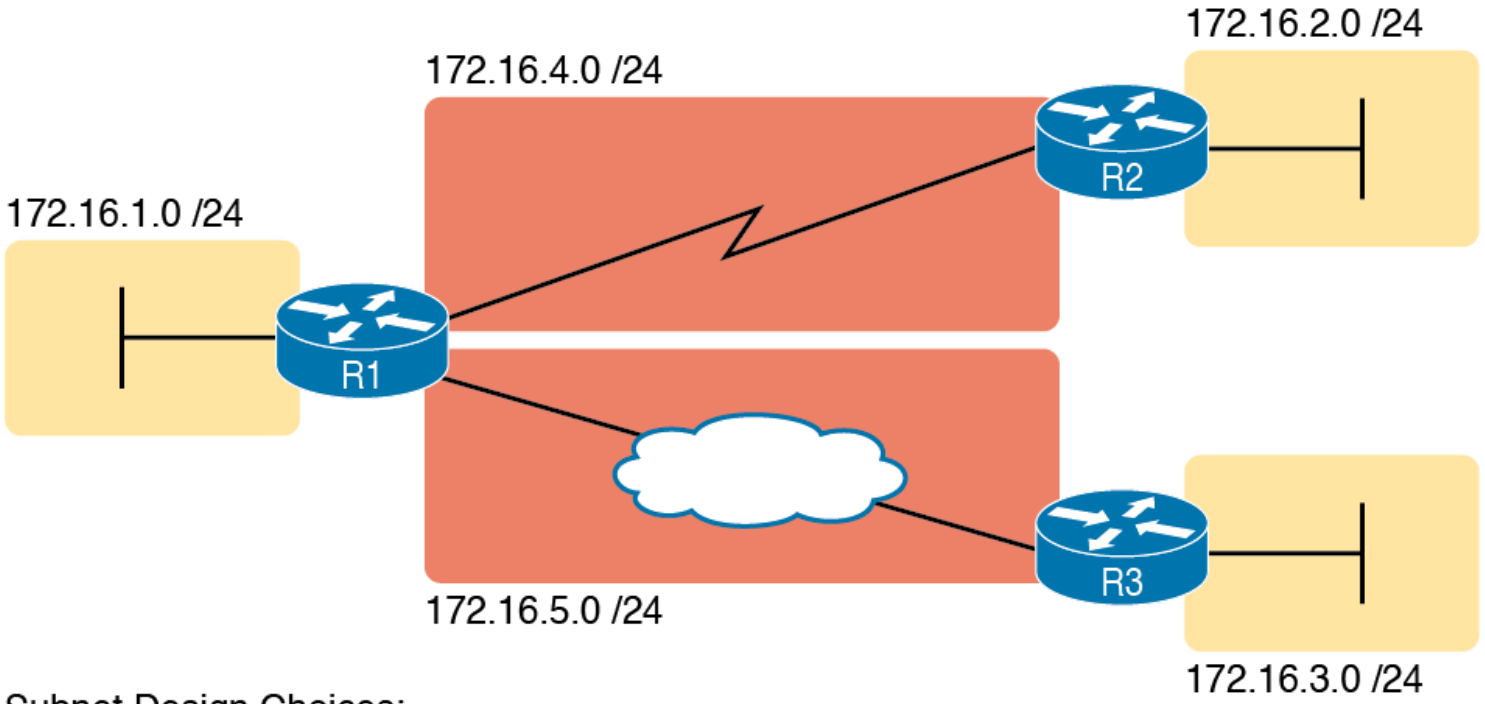
Creating the Subnet Mask—Binary—Class B Network



First Ten Subnets, Plus the Last Few, from 172.16.0.0, 255.255.255.0

Subnet Number	IP Addresses	Broadcast Address
172.16.0.0	172.16.0.1 – 172.16.0.254	172.16.0.255
172.16.1.0	172.16.1.1 – 172.16.1.254	172.16.1.255
172.16.2.0	172.16.2.1 – 172.16.2.254	172.16.2.255
172.16.3.0	172.16.3.1 – 172.16.3.254	172.16.3.255
172.16.4.0	172.16.4.1 – 172.16.4.254	172.16.4.255
172.16.5.0	172.16.5.1 – 172.16.5.254	172.16.5.255
172.16.6.0	172.16.6.1 – 172.16.6.254	172.16.6.255
172.16.7.0	172.16.7.1 – 172.16.7.254	172.16.7.255
172.16.8.0	172.16.8.1 – 172.16.8.254	172.16.8.255
172.16.9.0	172.16.9.1 – 172.16.9.254	172.16.9.255
----------Skipping many ----------		
172.16.254.0	172.16.254.1 – 172.16.254.254	172.16.254.255
172.16.255.0	172.16.255.1 – 172.16.255.254	172.16.255.255

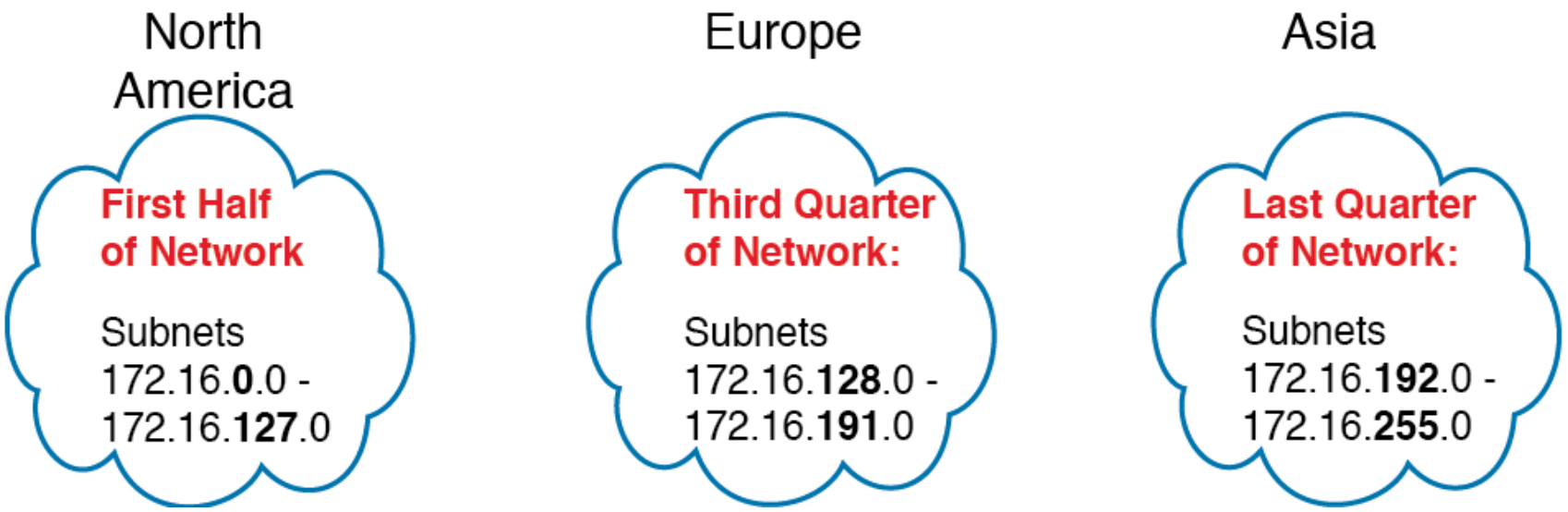
Example of Subnets Assigned to Different Locations



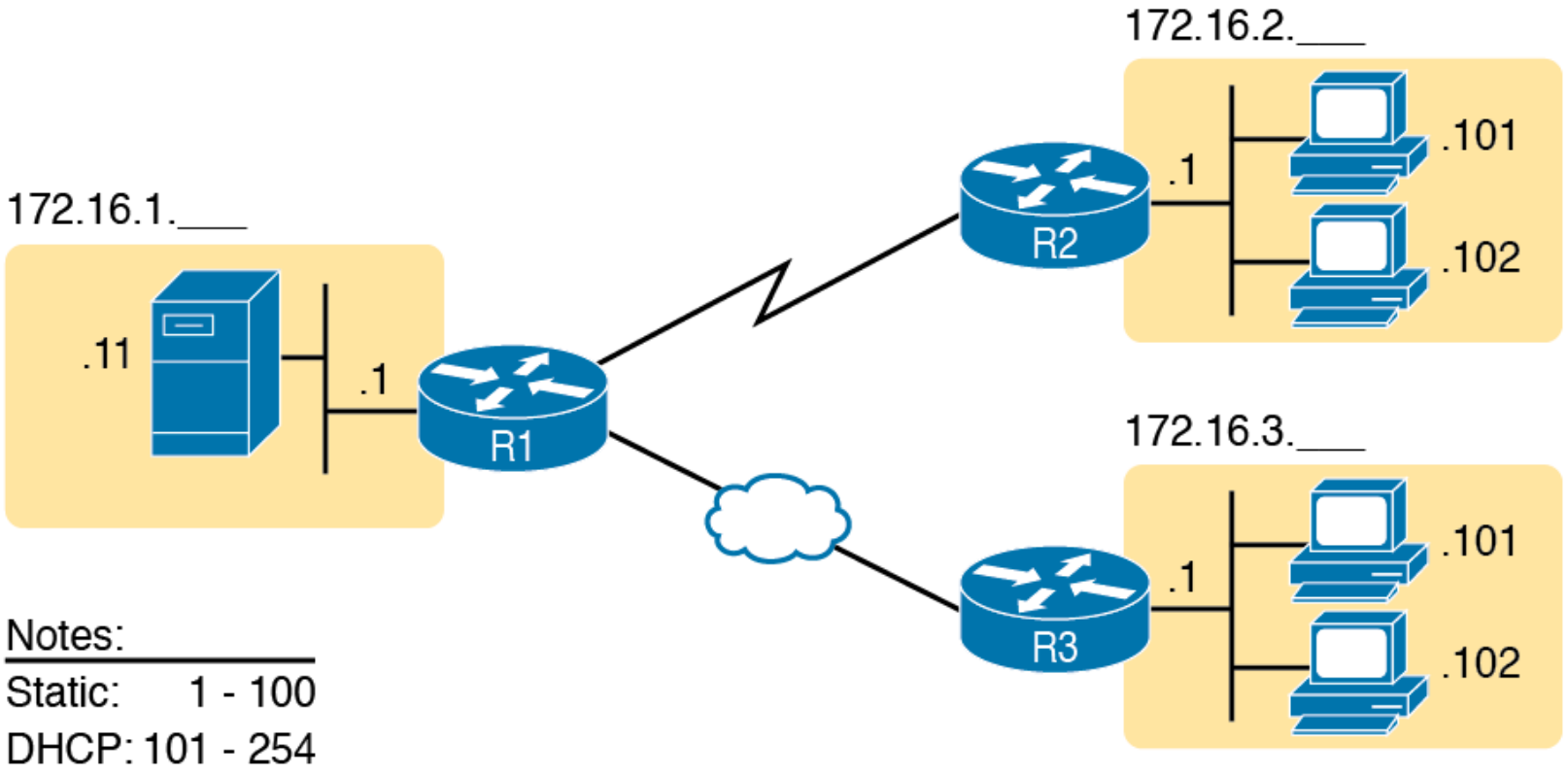
Subnet Design Choices:

Class B 172.16.0.0 /24 (255.255.255.0)

Reserving 50% of Subnets for North America and 25% Each for Europe and Asia



Static from the Low End and DHCP from the High End



Classful Network Concepts

- This chapter examines how to begin with a single IP address and then determine the following facts:
  - Class (A, B, or C)
  - Default mask
  - Number of network octets/bits
  - Number of host octets/bits
  - Number of host addresses in the network
  - Network ID
  - Network broadcast address
  - First and last usable address in the network

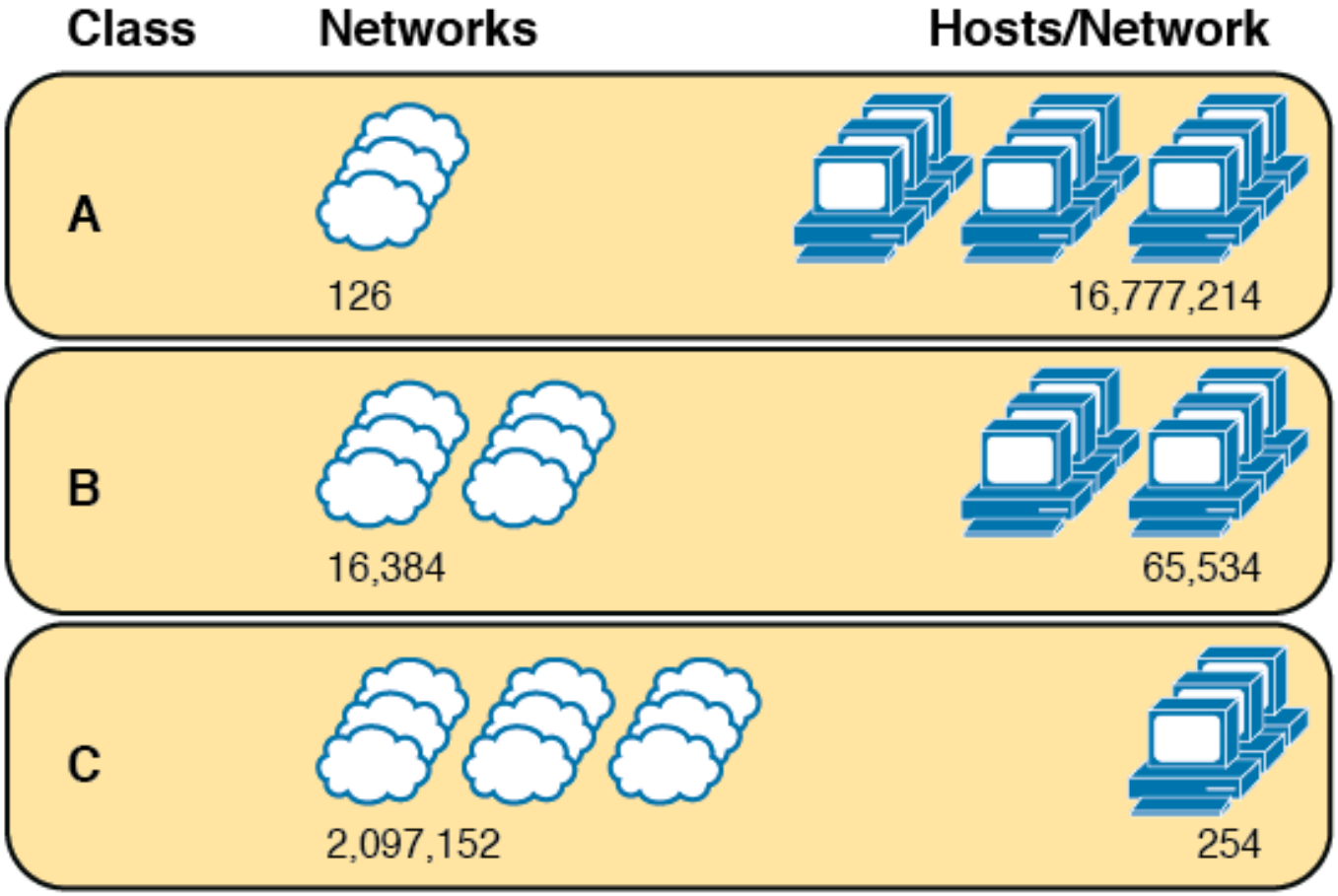
IPv4 Address Classes Based on First Octet Values

Class	First Octet Values	Purpose
A	1–126	Unicast (large networks)
B	128–191	Unicast (medium-sized networks)
C	192–223	Unicast (small networks)
D	224–239	Multicast
E	240–255	Reserved (formerly experimental)

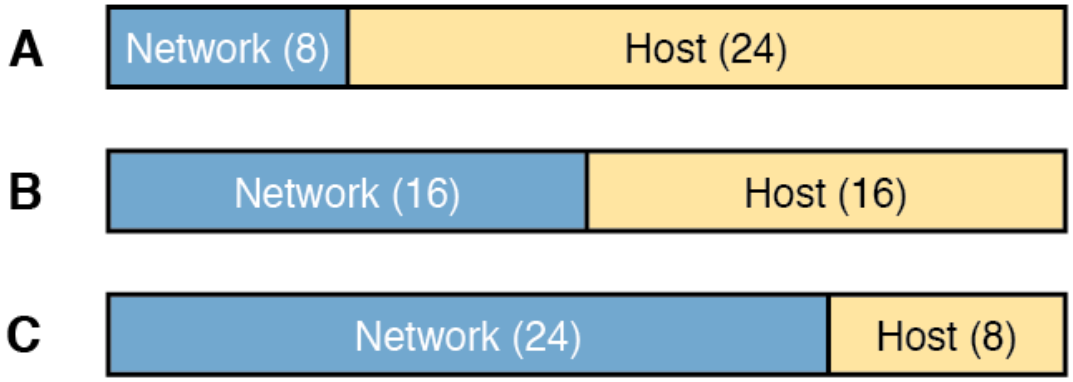
Key Facts for Classes A, B, and C

	Class A	**Class B**	**Class C**
First octet range	1 – 126	128 – 191	192 – 223
Valid network numbers	1.0.0.0 – 126.0.0.0	128.0.0.0 – 191.255.0.0	192.0.0.0 – 223.255.255.0
Total networks	$2^7 - 2 = 126$	$2^{14} = 16{,}384$	$2^{21} = 2{,}097{,}152$
Hosts per network	$2^{24} - 2$	$2^{16} - 2$	$2^8 - 2$
Octets (bits) in network part	1 (8)	2 (16)	3 (24)
Octets (bits) in host part	3 (24)	2 (16)	1 (8)
Default mask	255.0.0.0	255.255.0.0	255.255.255.0

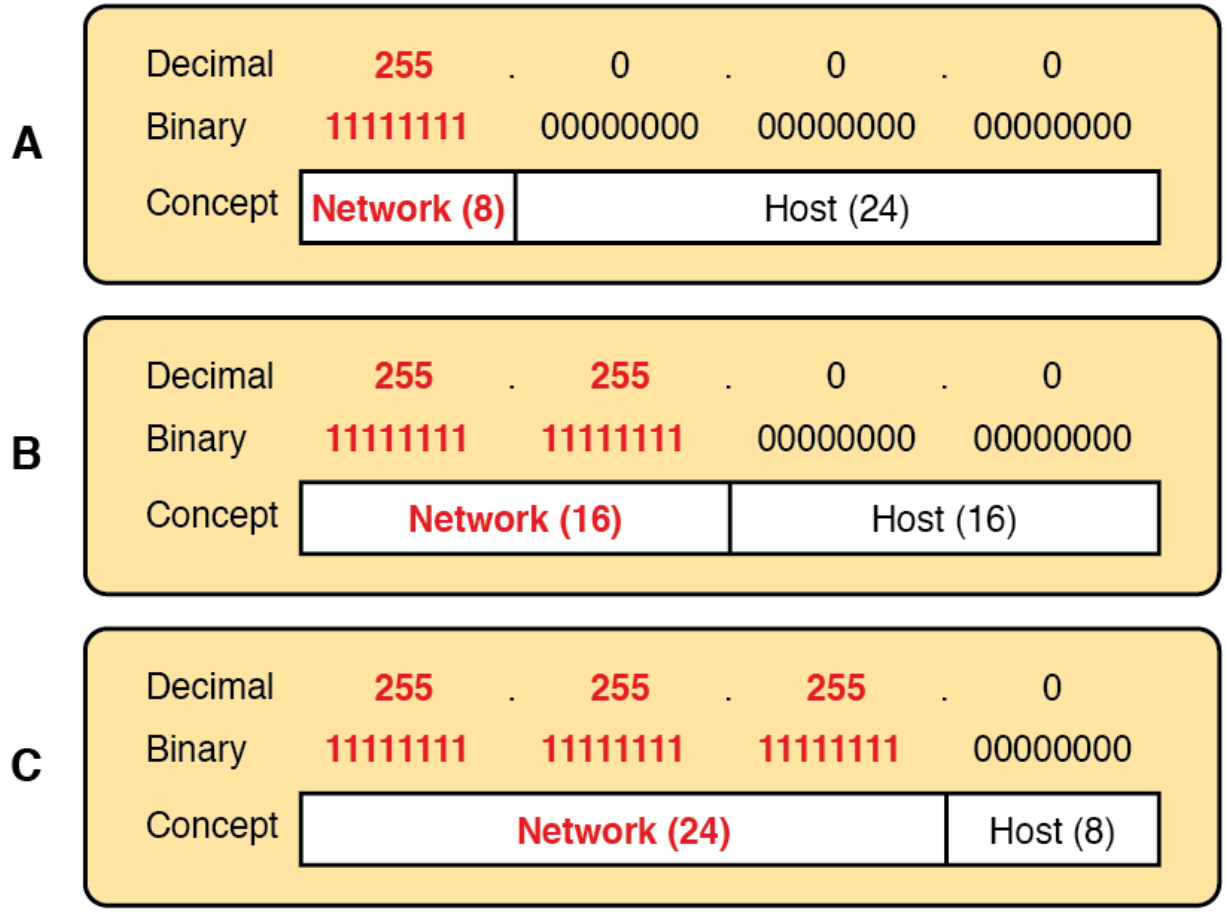
Numbers and Sizes of Class A, B, and C Networks



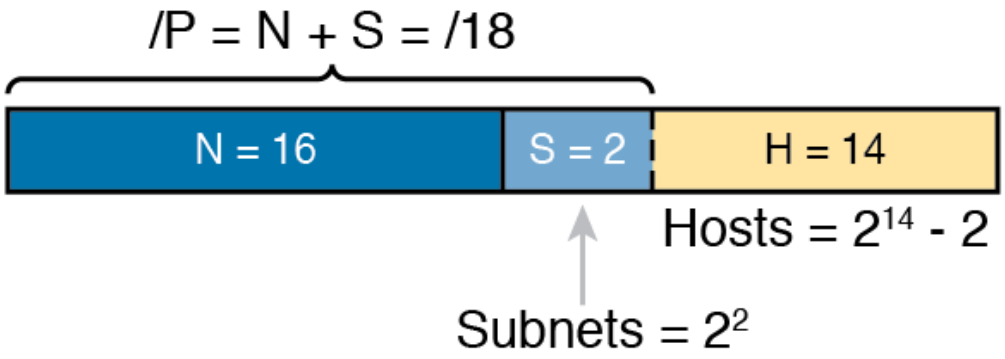
Sizes (Bits) of the Network and Host Parts of Unsubnetted Classful Networks



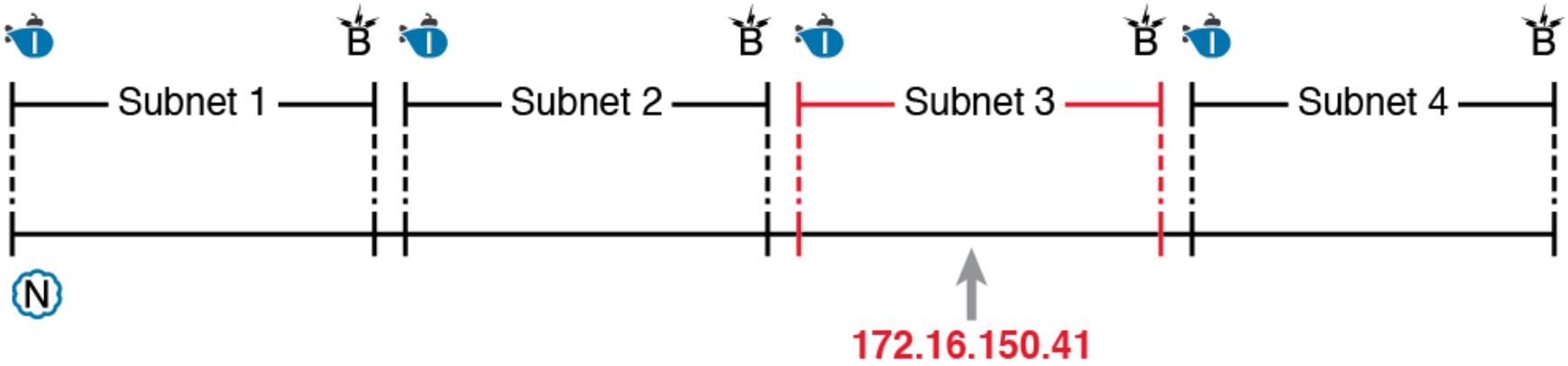
Default Masks for Classes A, B, and C



Address Structure: Class B Network, /18 Mask



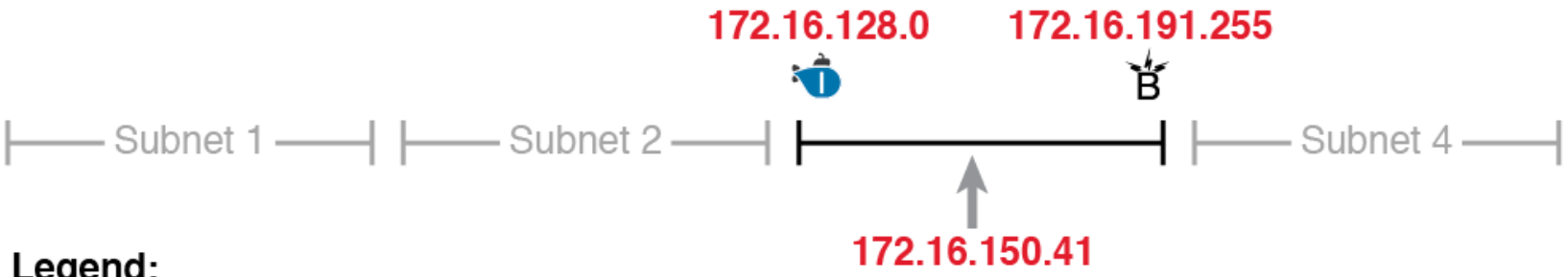
Network 172.16.0.0, Divided into Four Equal Subnets



Resident Subnet for 172.16.150.41, 255.255.192.0

10101100.00010000.10010110.00101001

11111111.11111111.11000000.00000000



Legend:

I Subnet ID

B Subnet Broadcast Address

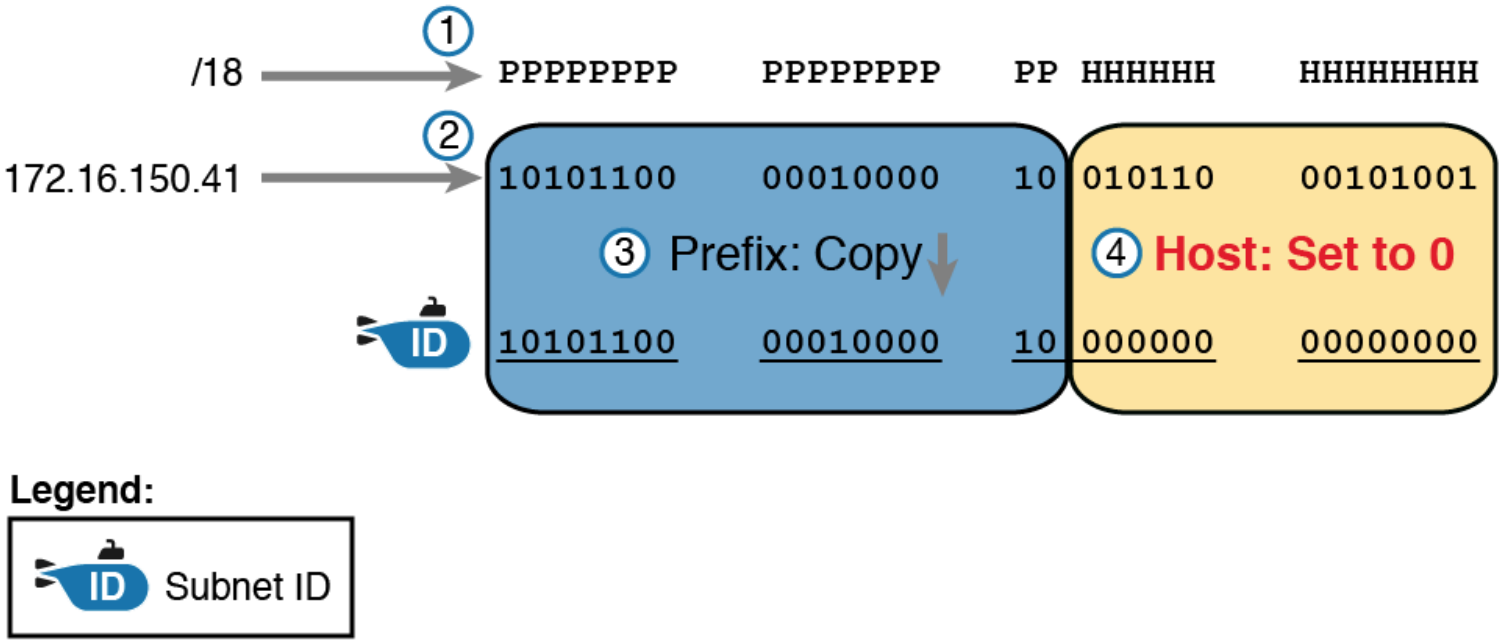
Summary of Subnet ID Key Facts

Definition	Number that represents the subnet
Numeric Value	First (smallest) number in the subnet
Literal Synonyms	Subnet number, subnet address, prefix, resident subnet
Common-Use Synonyms	Network, network ID, network number, network address
Typically Seen In…	Routing tables, documentation

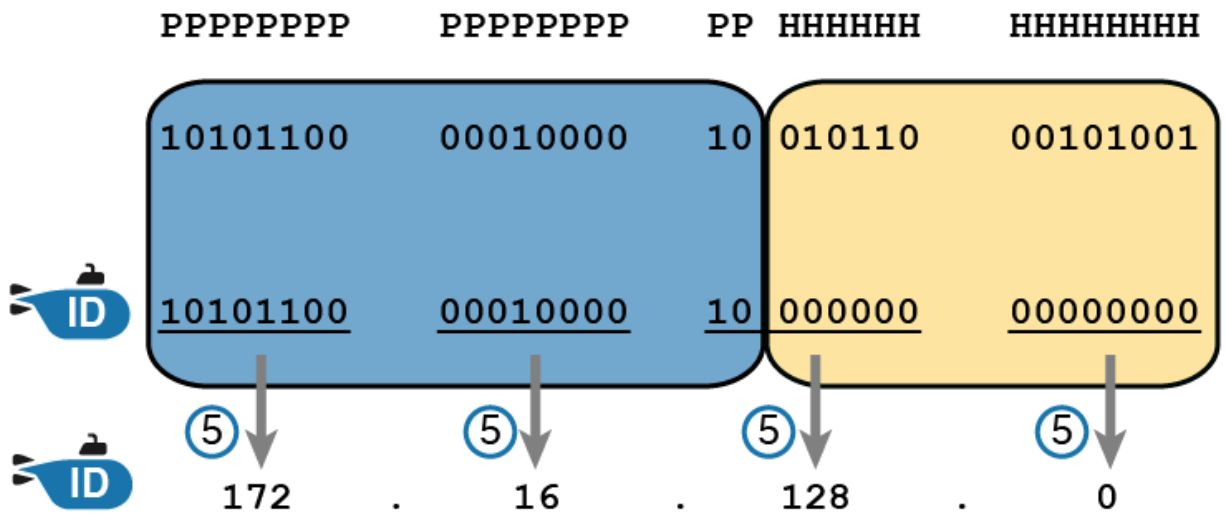
Summary of Subnet Broadcast Address Key Facts

Definition	A reserved number in each subnet that, when used as the destination address of a packet, causes the device to forward the packet to all hosts in that subnet
Numeric Value	Last (highest) number in the subnet
Literal Synonyms	Directed broadcast address
Broader-Use Synonyms	Network broadcast
Typically Seen In…	In calculations of the range of addresses in a subnet

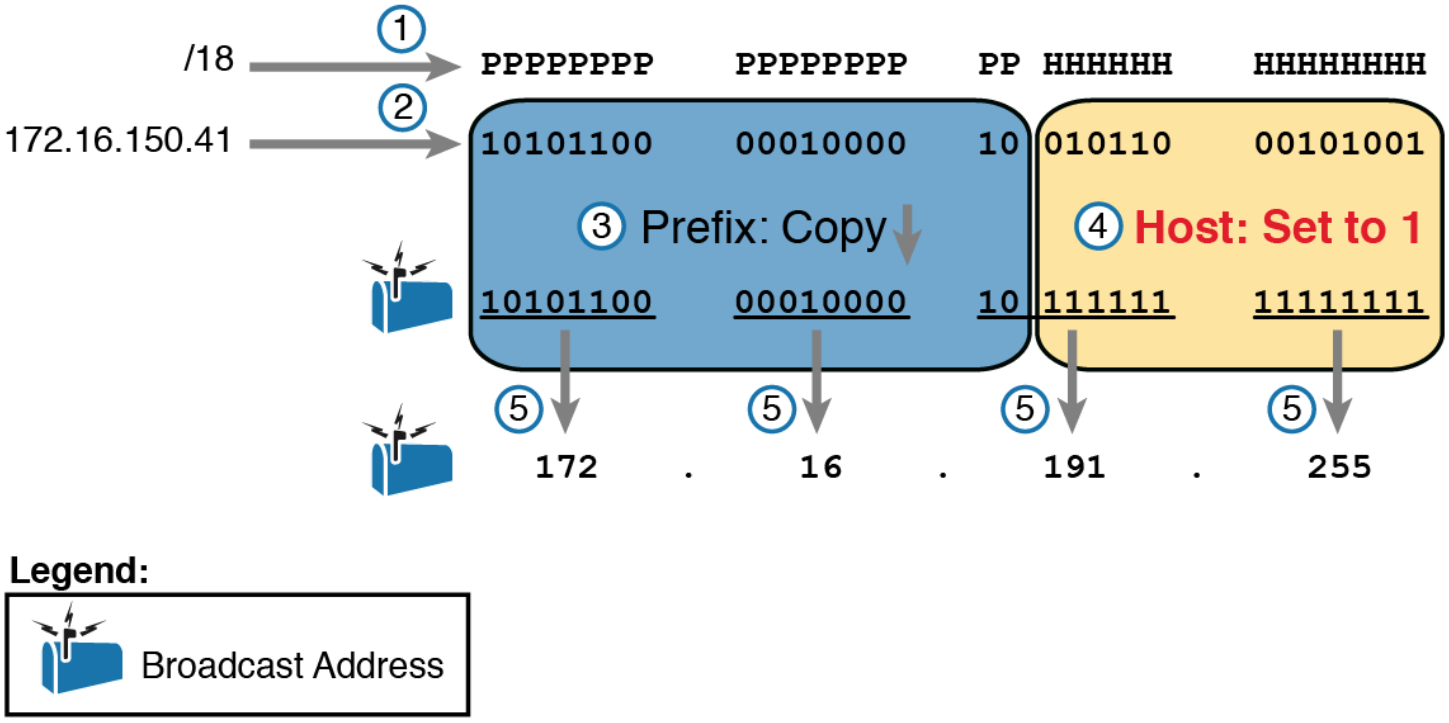
Binary Concept: Convert the IP Address to the Subnet ID



Converting the Subnet ID from Binary to DDN



Finding a Subnet Broadcast Address: Binary



Subnet Analysis for Address 8.1.4.5, Mask 255.255.0.0

Prefix Length	/16	11111111 11111111 **00000000 00000000**
Address	8.1.4.5	00001000 00000001 **00000100 00000101**
Subnet ID	8.1.0.0	00001000 00000001 **00000000 00000000**
Broadcast Address	8.1.255.255	00001000 00000001 **11111111 11111111**

Subnet Analysis for Subnet with Address 130.4.102.1, Mask 255.255.255.0

Prefix Length	/24	11111111 11111111 11111111 **00000000**
Address	130.4.102.1	10000010 00000100 01100110 **00000001**
Subnet ID	130.4.100.0	10000010 00000100 01100100 **00000000**
Broadcast Address	130.4.102.255	10000010 00000100 01100110 **11111111**

Subnet Analysis for Subnet with Address 199.1.1.100, Mask 255.255.255.0

Prefix Length	/24	11111111 11111111 11111111 **00000000**
Address	199.1.1.100	11000111 00000001 00000001 **01100100**
Subnet ID	199.1.1.0	11000111 00000001 00000001 **00000000**
Broadcast Address	199.1.1.255	11000111 00000001 00000001 **11111111**

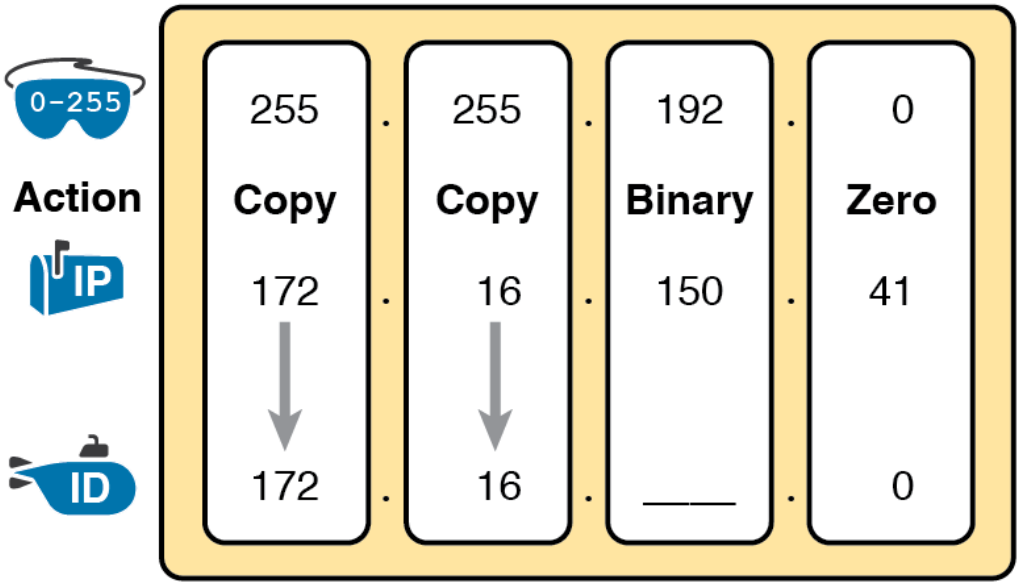
Subnet Analysis for Subnet with Address 130.4.102.1, Mask 255.255.252.0

Prefix Length	/22	11111111 11111111 11111**00 00000000**
Address	130.4.102.1	10000010 00000100 01100**110 00000001**
Subnet ID	130.4.100.0	10000010 00000100 01100**100 00000000**
Broadcast Address	130.4.103.255	10000010 00000100 01100**111 11111111**

Subnet Analysis for Subnet with Address 199.1.1.100, Mask 255.255.255.224

Prefix Length	/27	11111111 11111111 11111111 111**00000**
Address	199.1.1.100	11000111 00000001 00000001 011**00100**
Subnet ID	199.1.1.96	11000111 00000001 00000001 011**00000**
Broadcast Address	199.1.1.127	11000111 00000001 00000001 011**11111**

Binary Shortcut Example



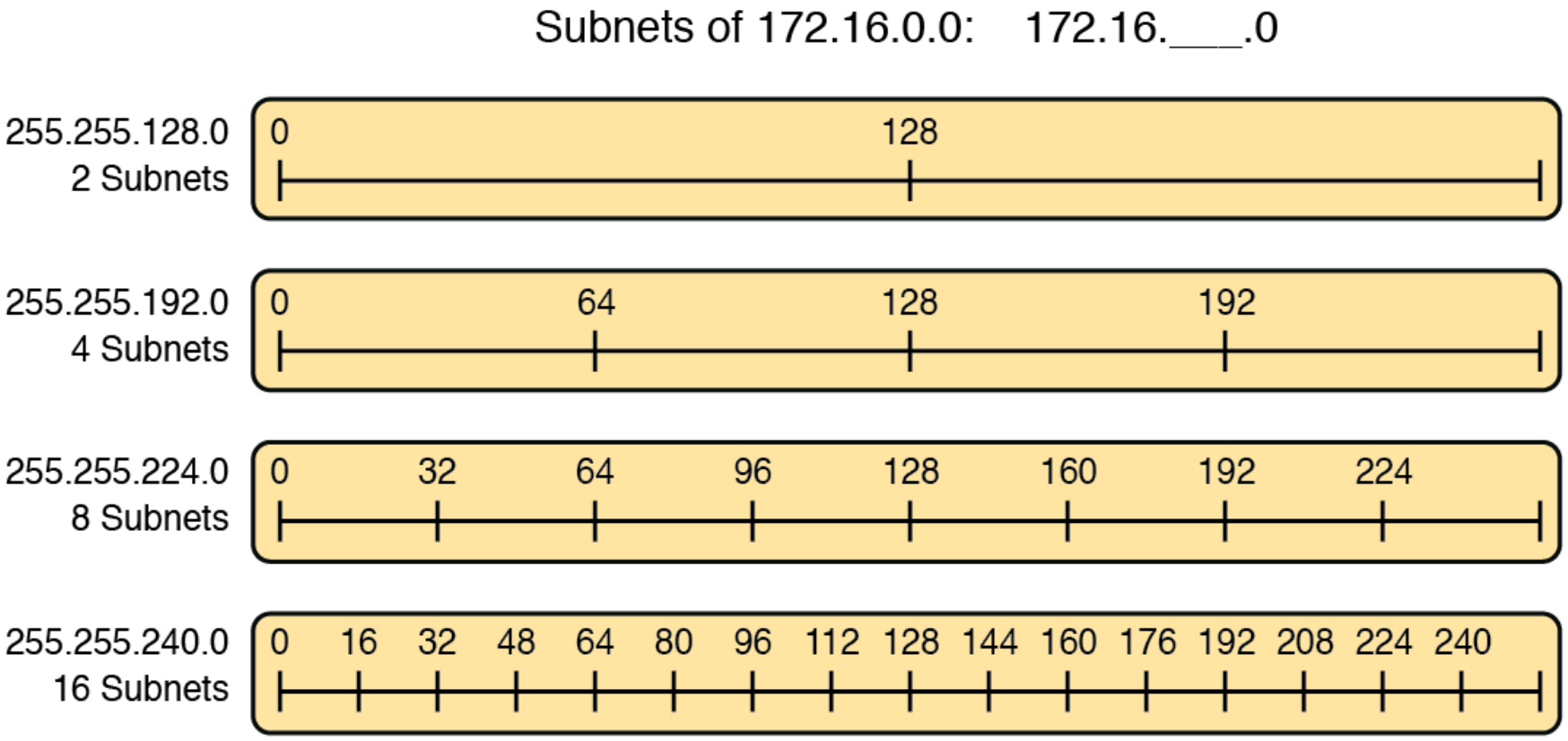
Legend:



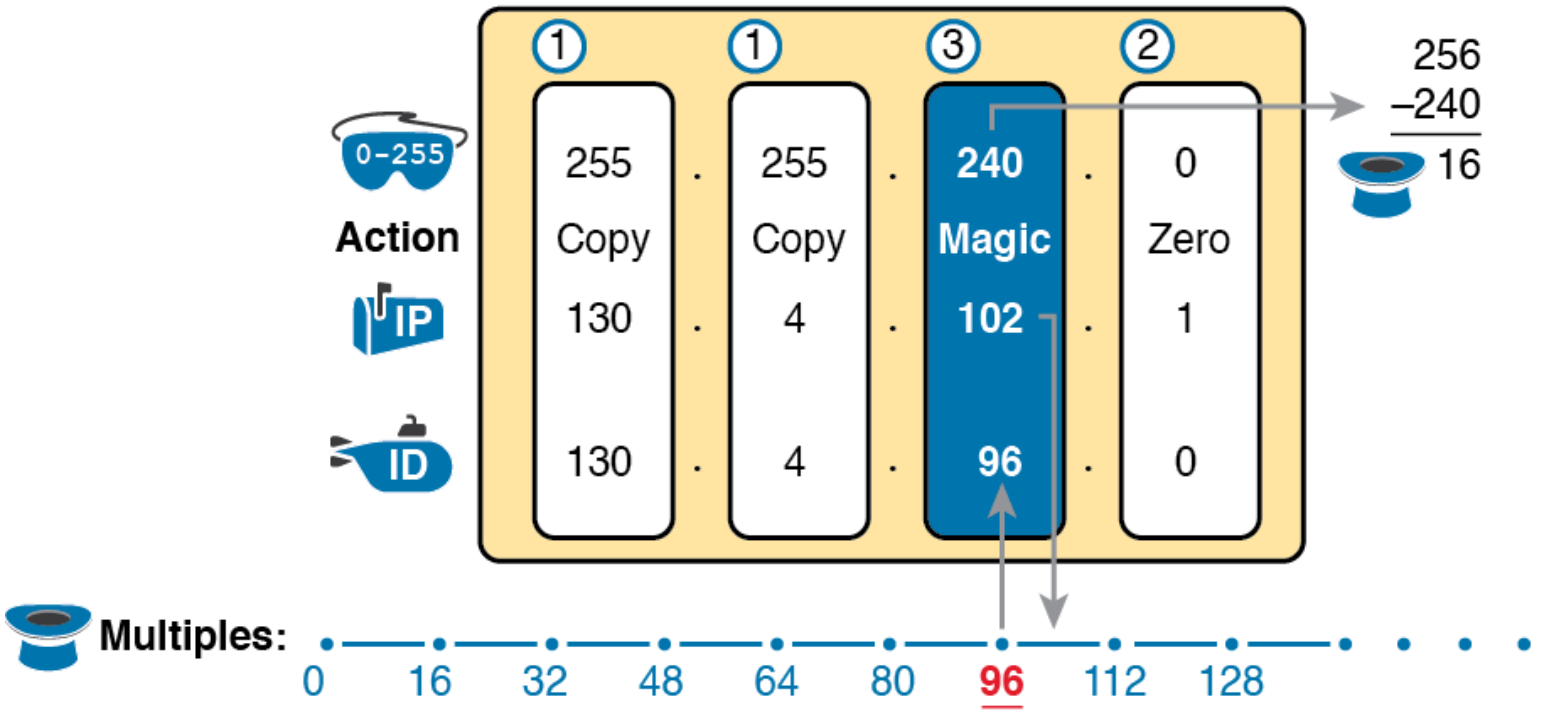
Practice Problems: Find Subnet ID and Broadcast, Easy Masks

	IP Address	Mask	Subnet ID	Broadcast Address
1	10.77.55.3	255.255.255.0		
2	172.30.99.4	255.255.255.0		
3	192.168.6.54	255.255.255.0		
4	10.77.3.14	255.255.0.0		
5	172.22.55.77	255.255.0.0		
6	1.99.53.76	255.0.0.0		

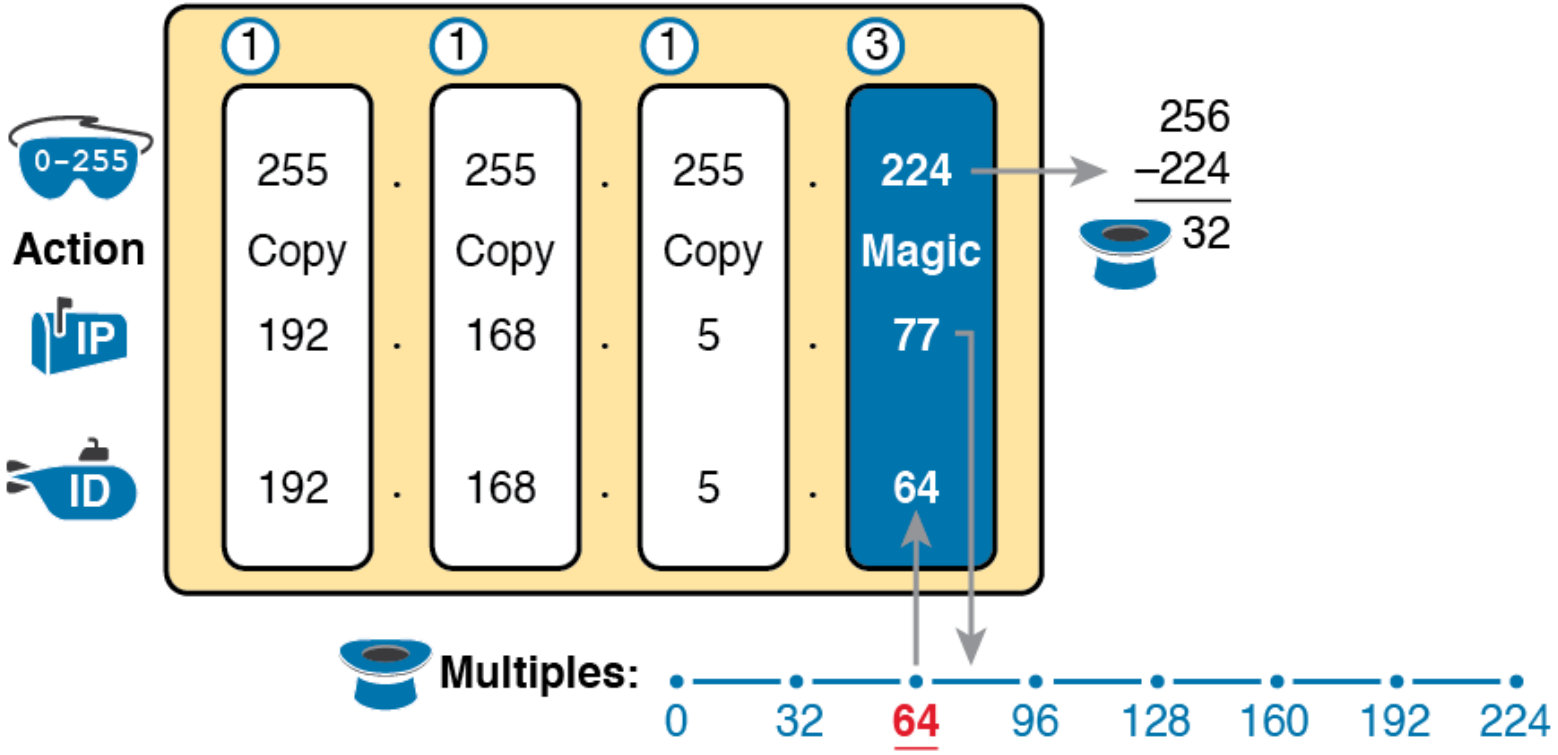
Numeric Patterns in the Interesting Octet



Find the Subnet ID: 130.4.102.1, 255.255.240.0



Resident Subnet for 192.168.5.77, 255.255.255.224



Practice Problems: Find Subnet ID, Difficult Masks

	IP Address	Mask	Subnet ID
1	10.77.55.3	255.248.0.0	
2	172.30.99.4	255.255.192.0	
3	192.168.6.54	255.255.255.252	
4	10.77.3.14	255.255.128.0	
5	172.22.55.77	255.255.254.0	
6	1.99.53.76	255.255.255.248	

Row 1, octet 2: 77 AND 248 → 01001101 AND 11111000 = 01001000 = **72**

Row 2, octet 3: 99 AND 192 → 01100011 AND 11000000 = 01000000 = **64**

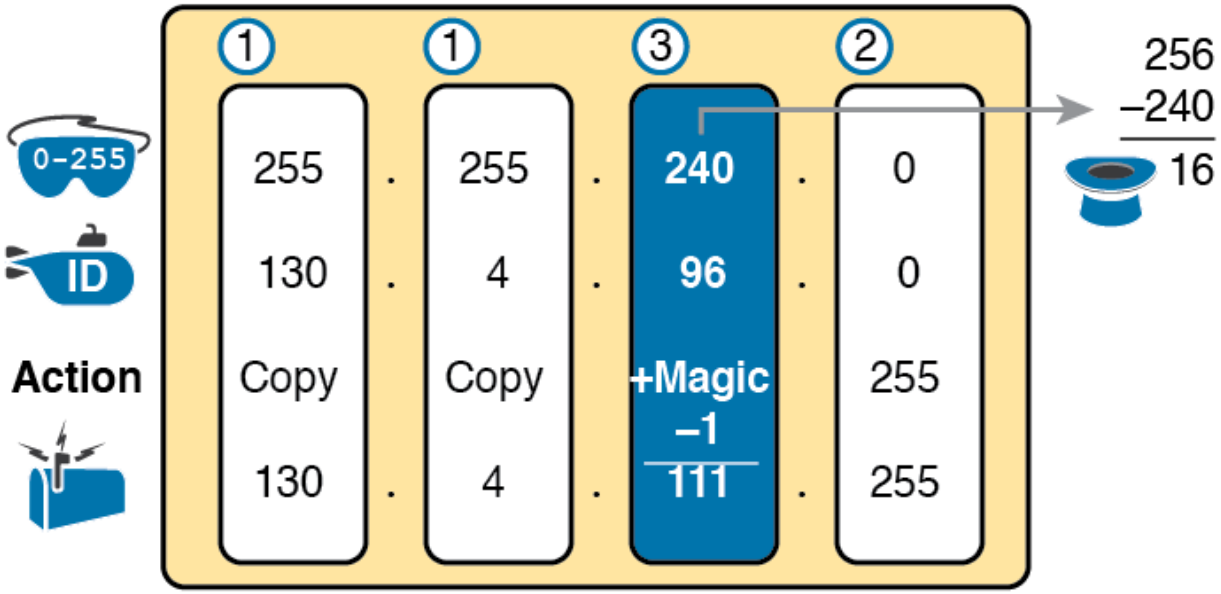
Row 3, octet 4: 54 AND 252 → 00110110 AND 11111100 = 00110100 = **52**

Row 4, octet 3 : 3 AND 128 → 00000011 AND 10000000 = 00000000 = **0**

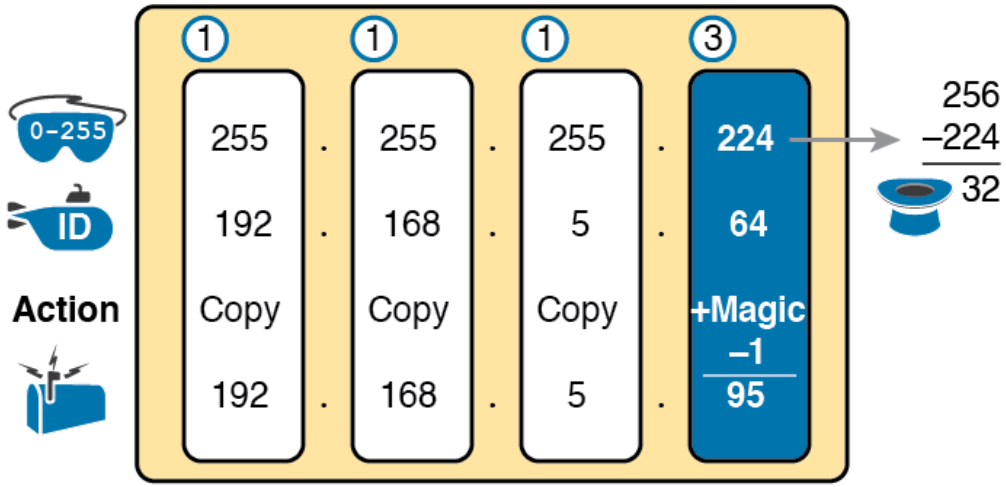
Row 5, octet 3: 55 AND 254 → 00110111 AND 11111110 = 00110110 = **54**

Row 6, octet 4: 76 AND 248 → 01001100 AND 11111000 = 01001000 = **72**

Find the Subnet Broadcast: 130.4.96.0, 255.255.240.0



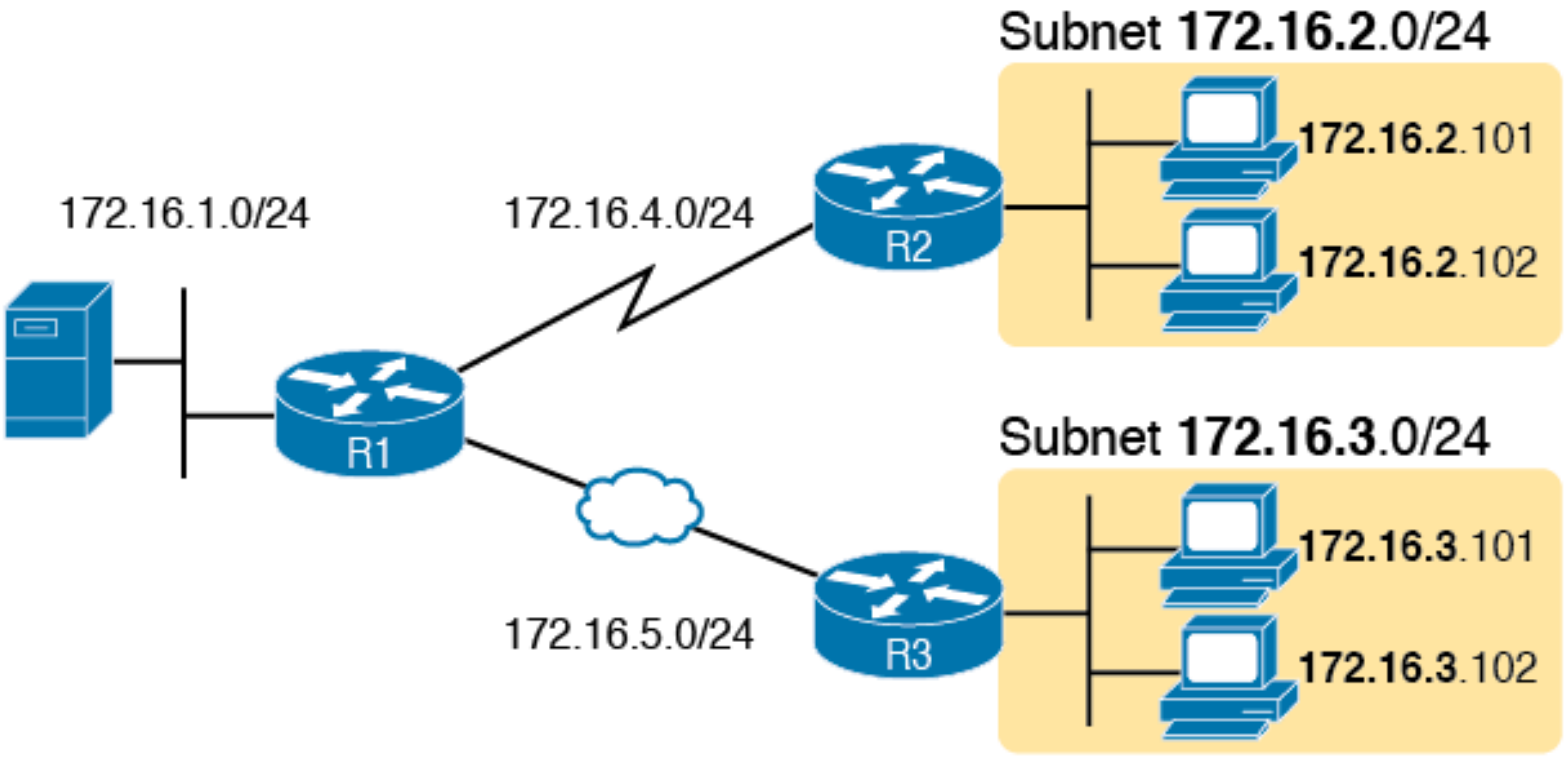
Find the Subnet Broadcast: 192.168.5.64, 255.255.255.224



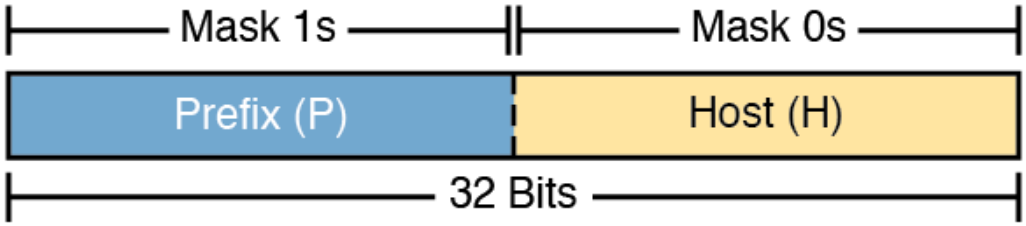
DDN Mask Values, Binary Equivalent, Magic Numbers, and Prefixes

Prefix, interesting octet 2	/9	/10	/11	/12	/13	/14	/15	/16
Prefix, interesting octet 3	/17	/18	/19	/20	/21	/22	/23	/24
Prefix, interesting octet 4	/25	/26	/27	/28	/29	/30		
Magic number	128	64	32	16	8	4	2	1
DDN mask in the interesting octet	128	192	224	240	248	252	254	255

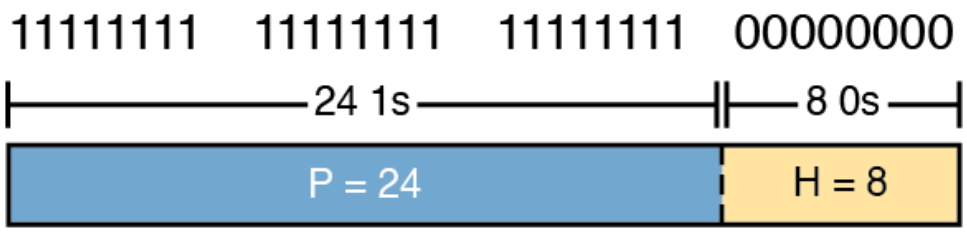
Simple Subnet Design, with Mask /24



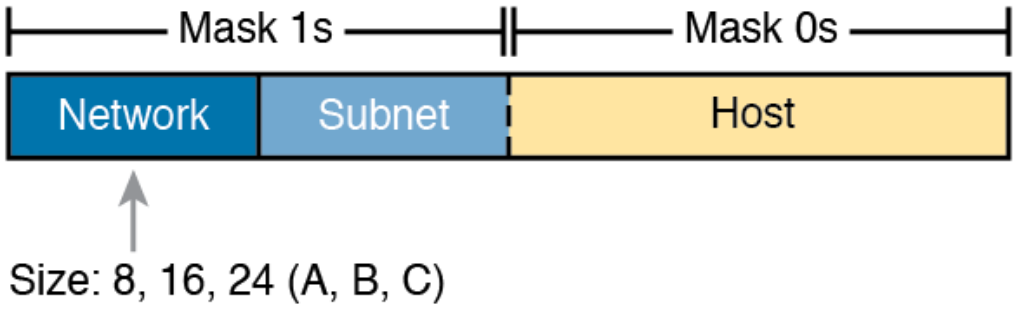
Prefix (Subnet) and Host Parts Defined by Mask 1s and 0s



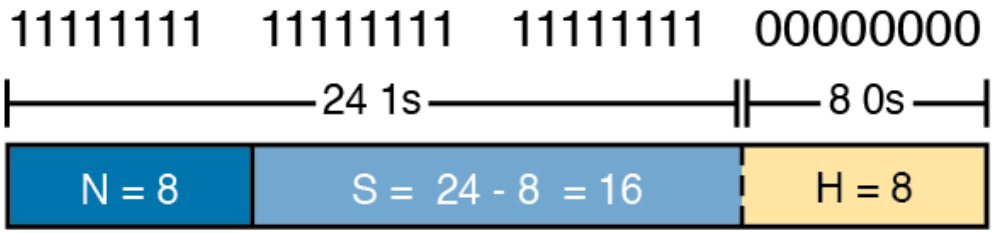
Mask 255.255.255.0: P=24, H=8



Class Concepts Applied to Create Three Parts

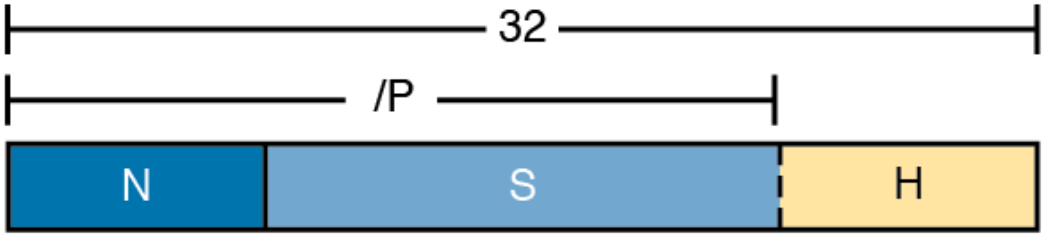


Subnet 10.1.1.0, Mask 255.255.255.0: N=8, S=16, H=8



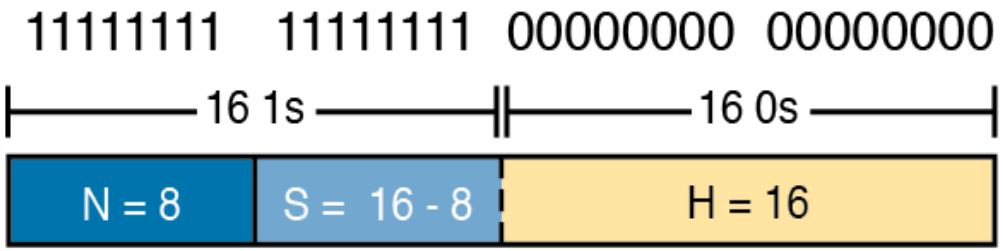
Based on Class

Relationship Between /P, N, S, and H



Class:
A: N = 8
B: N = 16
C: N = 24

Visual Representation of Problem: 8.1.4.5, 255.255.0.0



Example – 200.1.1.1 255.255.255.252

- **Step 1** 255.255.255.252 = /30, so P=30.
- **Step 2** 200.1.1.1 is in the range 192–223 in the first octet, so it is Class C; so N=24.
- **Step 3** S = P - N = 30 - 24 = 6.
- **Step 4** H = 32 - P = 32 – 30 = 2.
- **Step 5** 2^2 - 2 = 2 hosts/subnet
- **Step 6** 2^6 = 64 subnets.